

CS & IT ENGINEERING

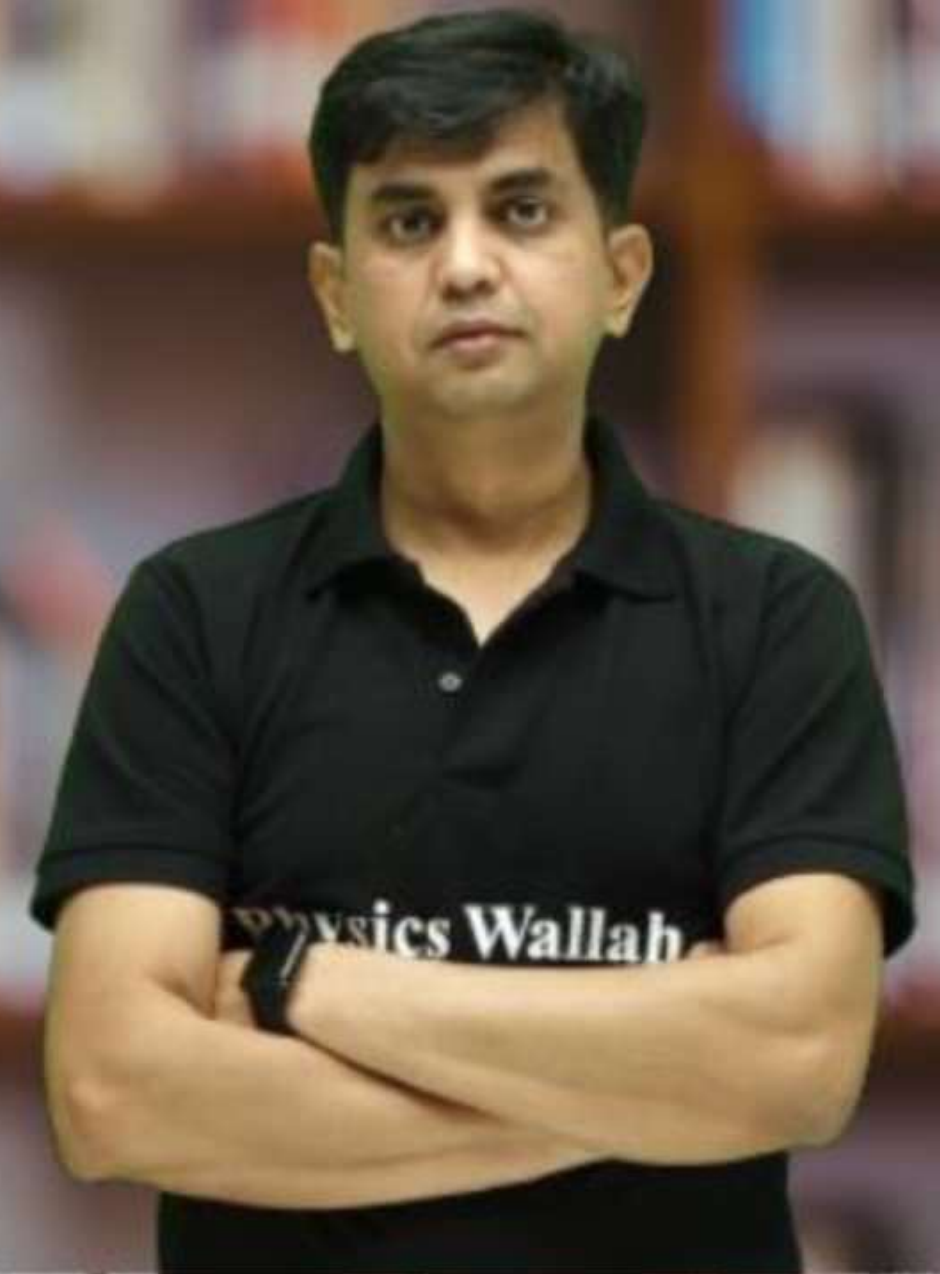


Computer Network

IPv4 Header

Lecture No. - 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

Fragmentation at Source Host

Topic

Fragmentation at Router





Topics to be Covered



Topic

Fragmentation & Reassembly



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
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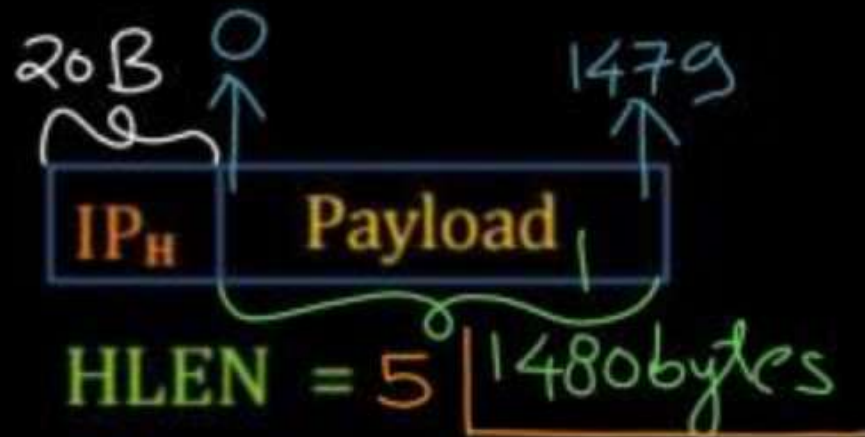
Example 8 :-

[NAT]



#Q. Consider following IPv4 datagrams carrying one UDP segment (created in example 7) are arrived at intermediate IPv4 router. The next network (link) MTU at router is 500 bytes, calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation ?

Ans = 11



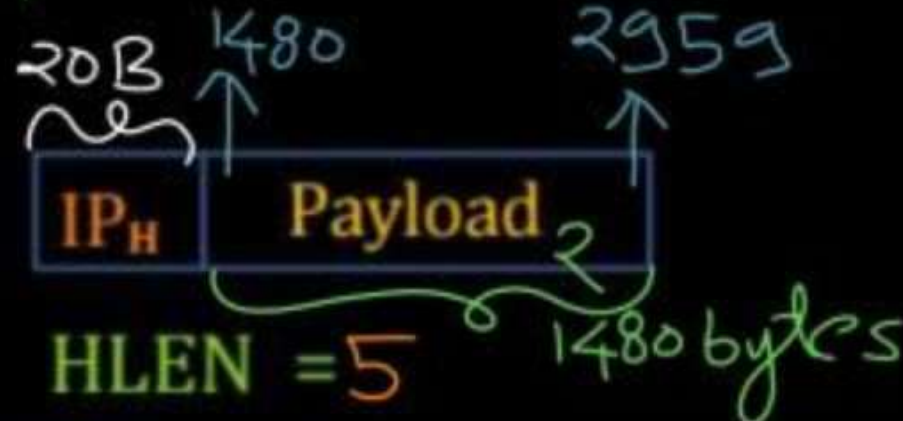
HLEN = 5 | 1480 bytes

TL = 1500

Id No. = Y

Offset = $\frac{0}{8} = 0$

MF bit = 1



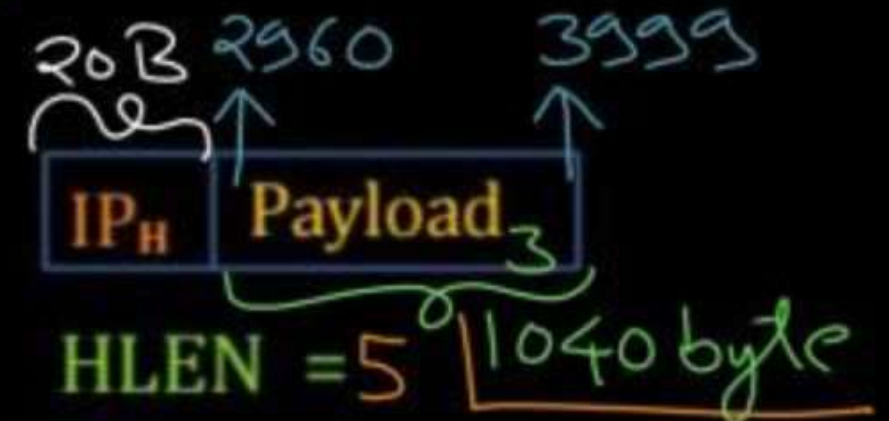
HLEN = 5 | 2980 bytes

TL = 1500

Id No. = Y

Offset = $\frac{1480}{8} = 185$

MF bit = 1



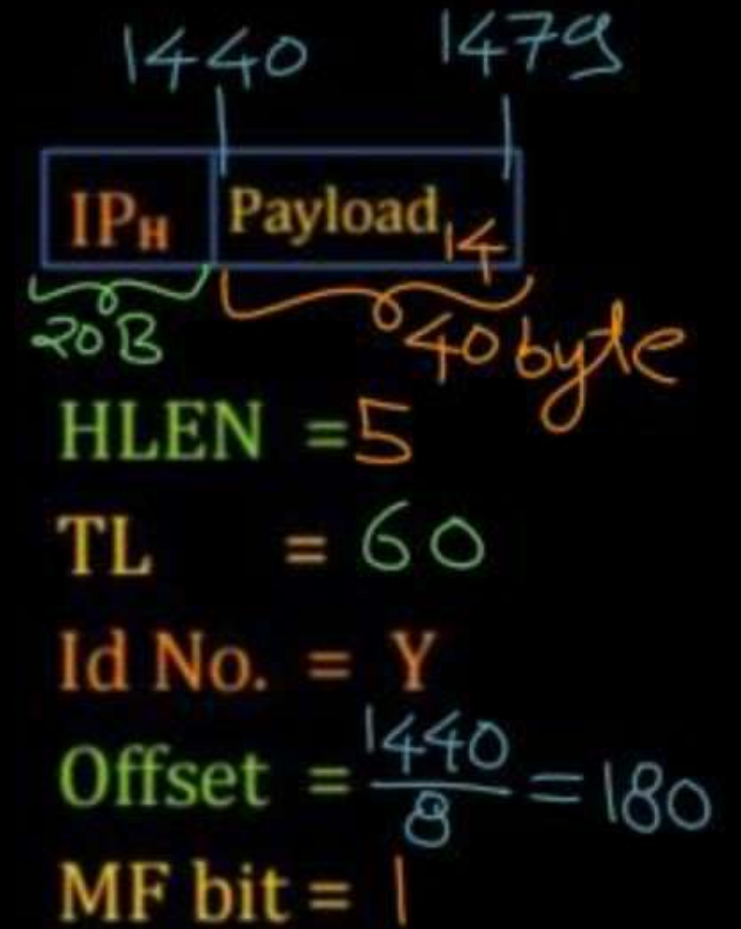
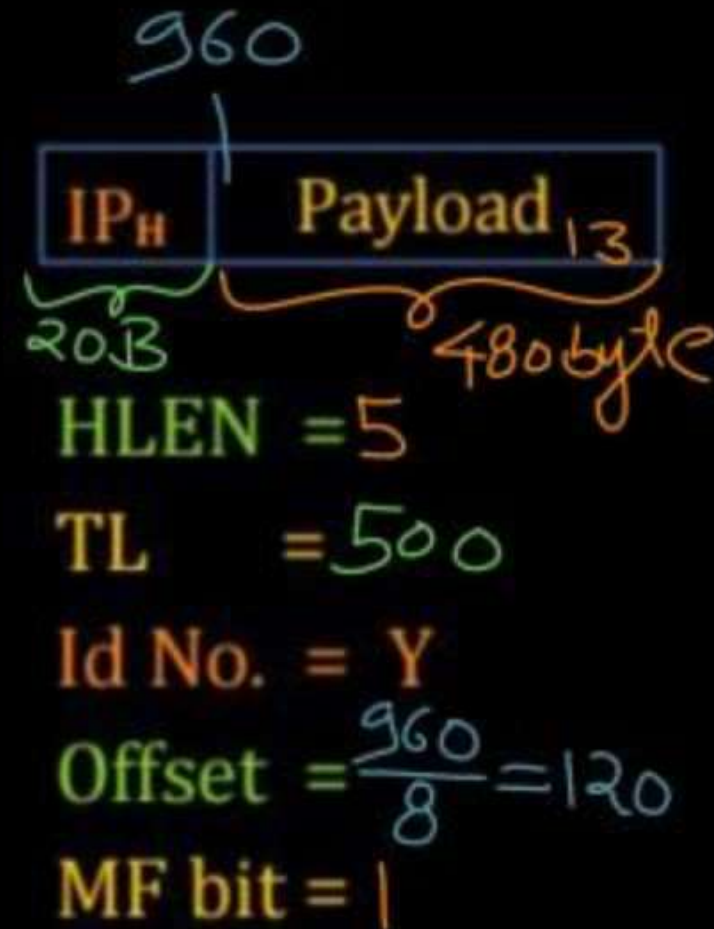
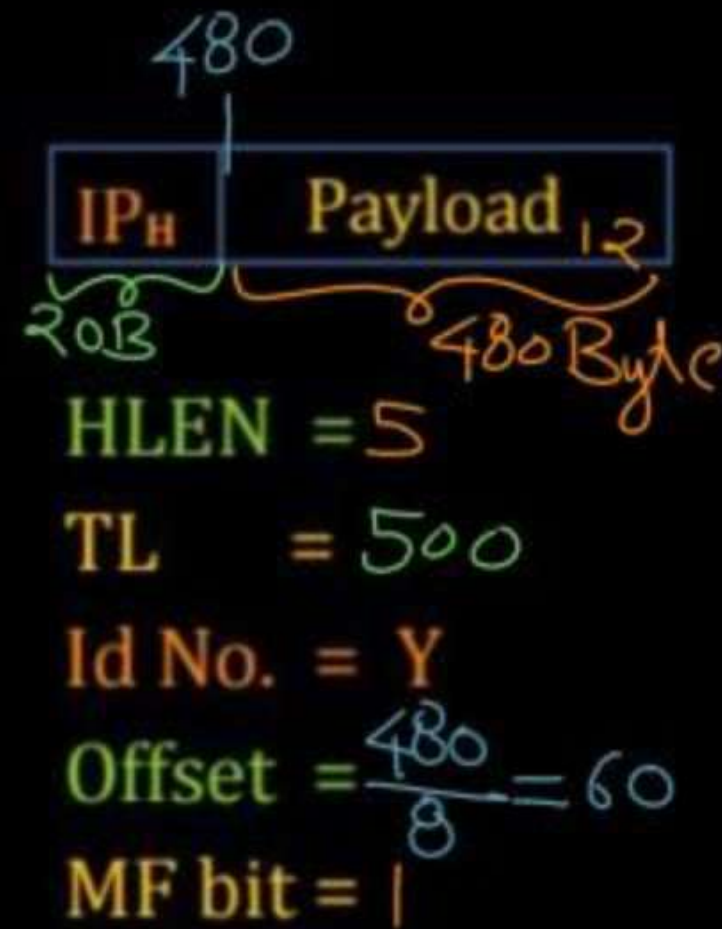
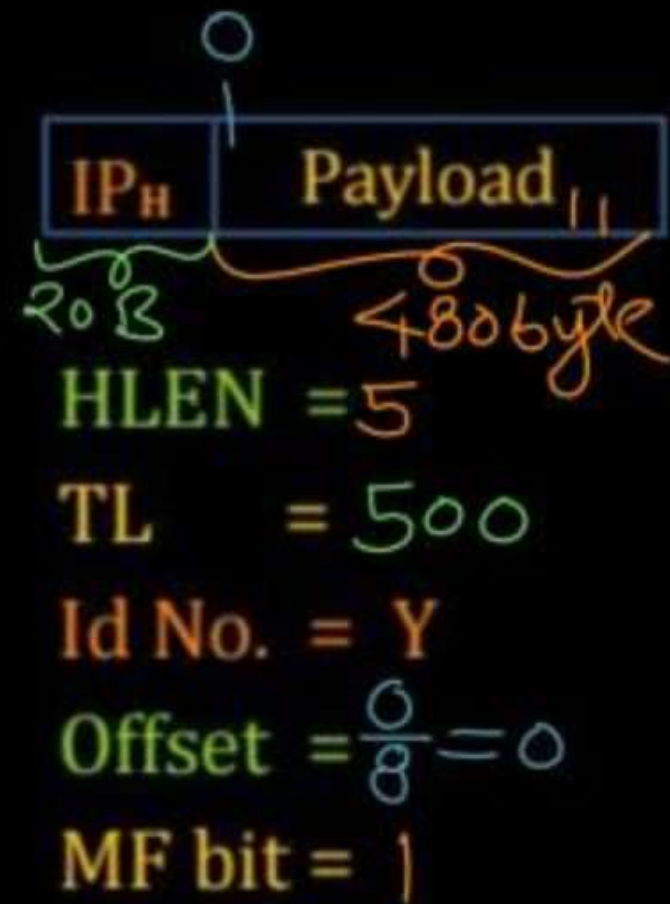
HLEN = 5 | 4020 bytes

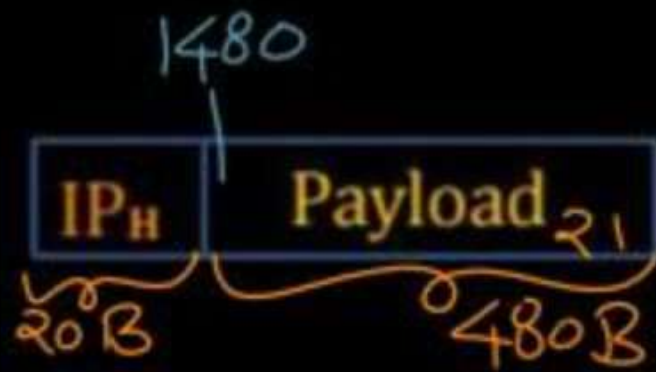
TL = 1060

Id No. = Y

Offset = $\frac{2960}{8} = 370$

MF bit = 0





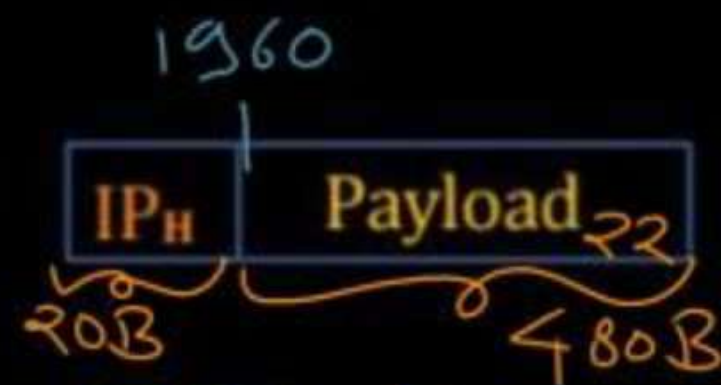
HLEN = 5

TL = 500

Id No. = Y

Offset = $\frac{1480}{8} = 185$

MF bit = 1



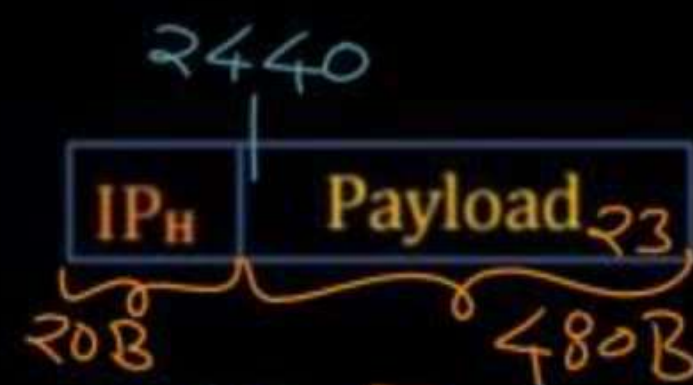
HLEN = 5

TL = 500

Id No. = Y

Offset = $\frac{1960}{8} = 245$

MF bit = 1



HLEN = 5

TL = 500

Id No. = Y

Offset = $\frac{2440}{8} = 305$

MF bit = 1



HLEN = 5

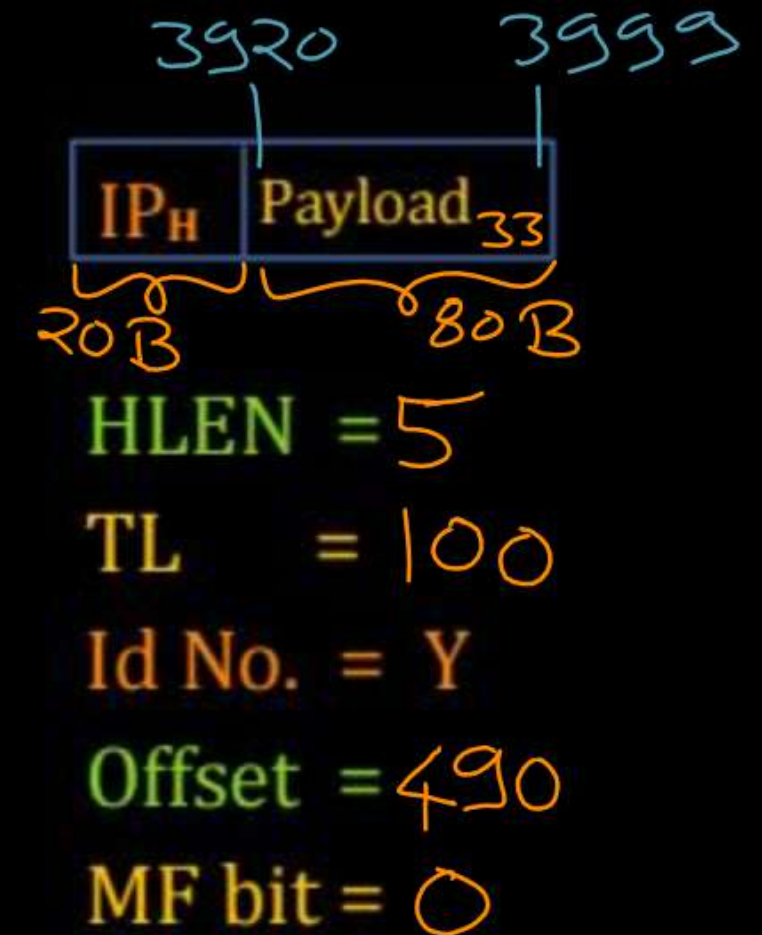
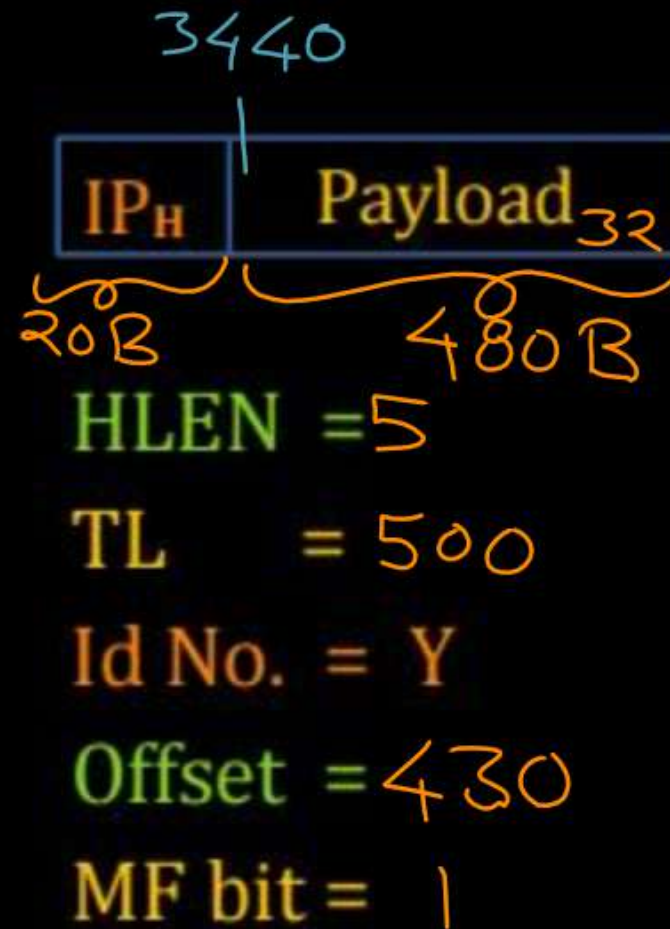
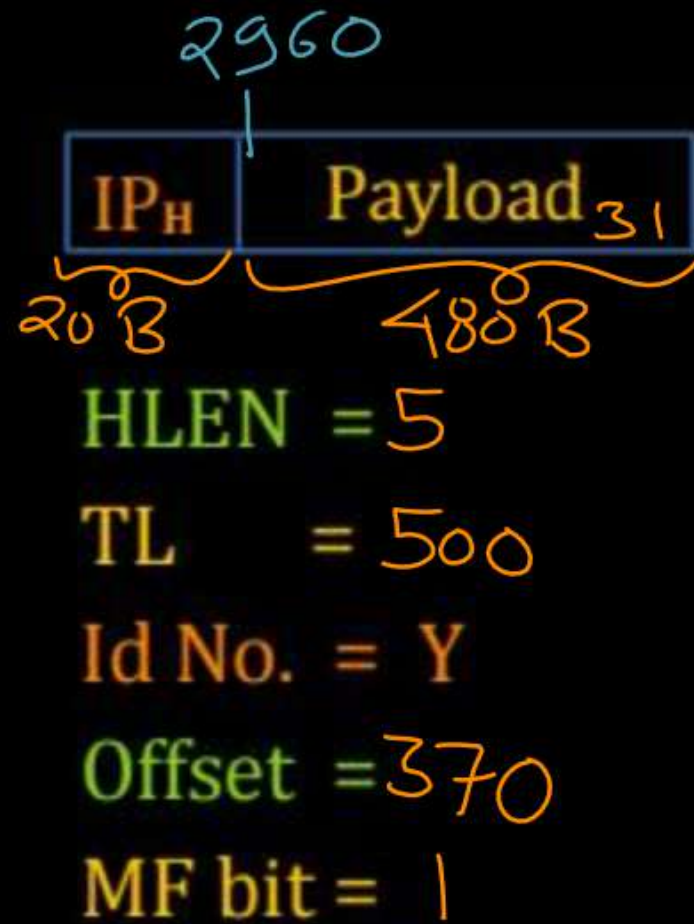
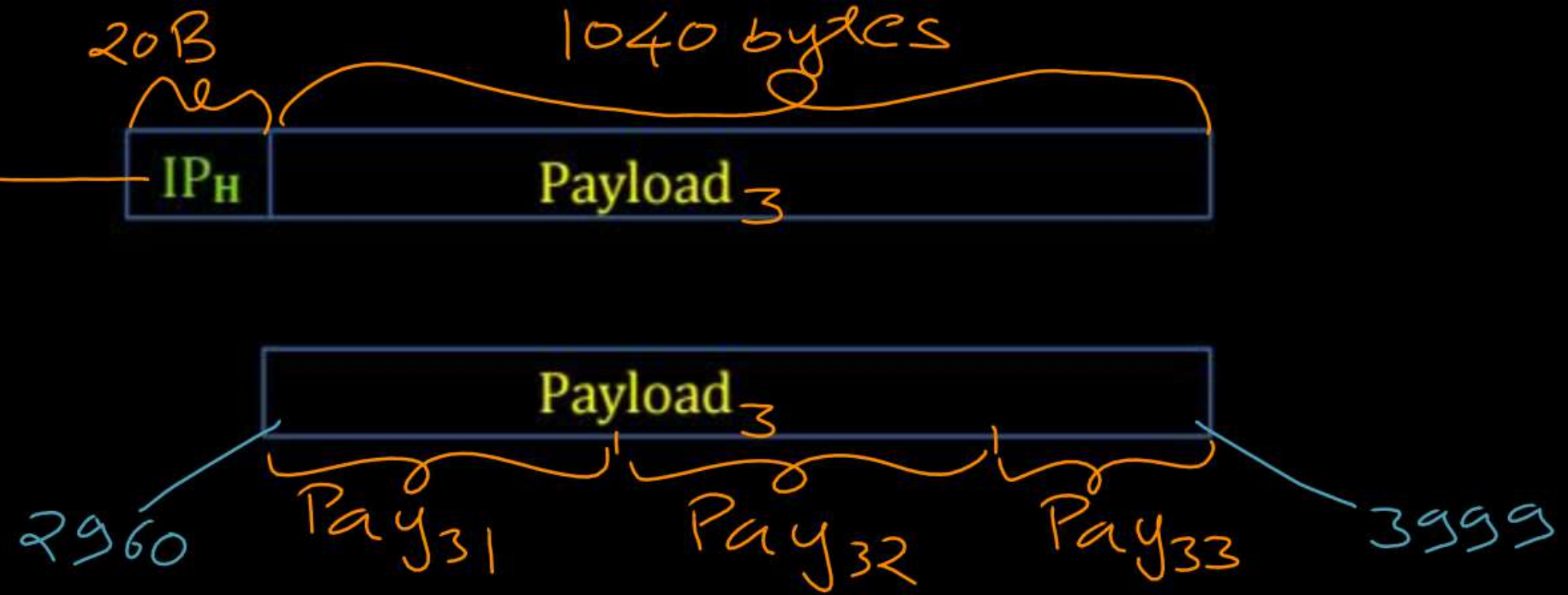
TL = 60

Id No. = Y

Offset = $\frac{2920}{8} = 365$

MF bit = 1

HLEN = 5
 TL = 1060
 Id No. = Y
 Offset = 370
 MF bit = 0



$$\text{HLEN} = 5$$

$$\text{Total Length} = 1060$$

$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [1060 - (5 * 4)] \text{ bytes} \\ &= 1040 \text{ bytes} \end{aligned}$$

$$\text{MTU} = 500 \text{ bytes}$$

$$\begin{aligned} \text{New Payload Size} &= [\text{MTU} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [500 - (5 * 4)] \text{ bytes} \\ &= 480 \end{aligned}$$

$$\text{Total Number of IP fragments } \underset{\uparrow}{[N]} = \left\lceil \frac{\text{Old Payload Size}}{\text{New Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{1040 \text{ bytes}}{480 \text{ bytes}} \right\rceil$$

$$= \lceil 2.16 \rceil$$

$$N = 3$$

Total length of the last IP fragment =

$$[\text{HLEN} * 4] + [\text{Old Payload Size} - (N-1) * \text{New Payload Size}] \text{ bytes}$$

$$= (5 * 4) + [1040 - (3-1) * 480] \text{ bytes}$$

$$= (20 + 80) \text{ bytes}$$

$$= 100 \text{ bytes}$$

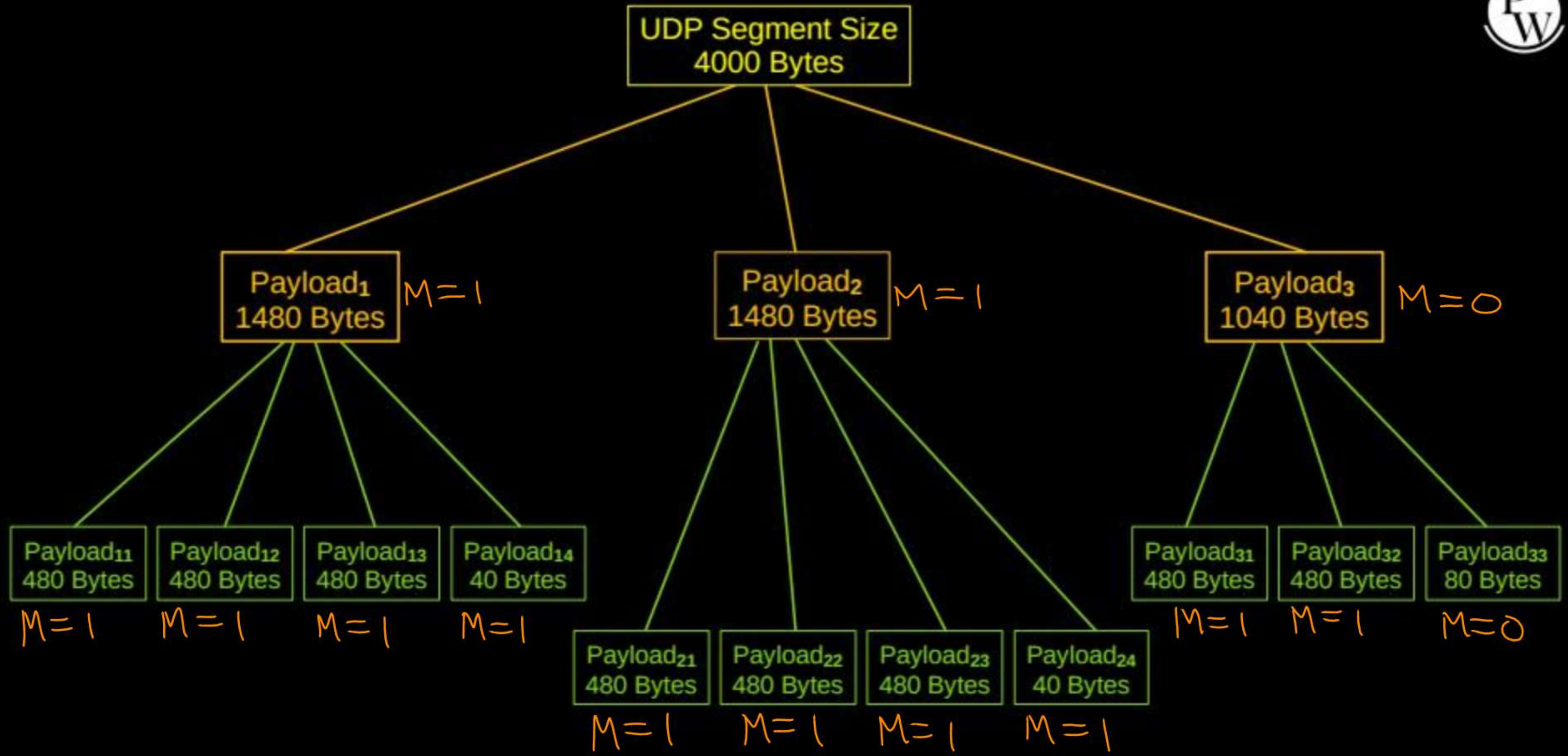
$$\text{Offset value of the last IP fragment} = \text{Old offset value} + \left[\frac{(N-1) * \text{New Payload Size}}{8} \right]$$

$$= 370 + \left[\frac{(3-1) * 480}{8} \right]$$

$$= 370 + \left[\frac{960}{8} \right]$$

$$= 370 + 120$$

$$= 490$$



[NAT]

[GATE-2016] [2 Mark]



#Q. An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is _____.

Ans = 13

$$\begin{aligned} \text{Total Length} &= \underline{1000 \text{ bytes}} \\ \text{Header Size} &= \underline{20 \text{ bytes}} \end{aligned}$$

$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - \text{Header Size}] \text{ bytes} \\ &= (1000 - 20) \text{ bytes} \\ &= 980 \text{ bytes} \end{aligned}$$

$$\text{MTU} = \underline{100 \text{ bytes}}$$

$$\begin{aligned} \text{New Payload Size} &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (100 - 20) \text{ bytes} \\ &= 80 \text{ bytes} \end{aligned}$$

$$\begin{aligned}
 \text{Total Number of IP fragments [N]} &= \left\lceil \frac{\text{Old Payload Size}}{\text{New Payload Size}} \right\rceil \\
 &= \left\lceil \frac{980 \text{ bytes}}{80 \text{ bytes}} \right\rceil \\
 &= 13
 \end{aligned}$$

[MCQ]

[GATE-2014] [2 Mark]



#Q. An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are



- ☒ A MF bit: 0, Datagram Length: 1444; Offset: 370
- ☐ B MF bit: 1, Datagram Length: 1424; Offset: 185
- ☐ C MF bit: 1, Datagram Length: 1500; Offset: 37
- ☐ D MF bit: 0, Datagram Length: 1424; Offset: 2960

Ans: A

Total Length = [4404 bytes]

Header Size = [20 bytes]

Old Payload Size = [Total Length - Header Size] bytes

$$= (4404 - 20) \text{ bytes}$$

$$= 4384 \text{ bytes}$$

MTU = 1500 bytes

New Payload Size = [MTU - Header Size] bytes

$$= (1500 - 20) \text{ bytes}$$

$$= 1480 \text{ bytes}$$

$$\text{Total Number of IP fragments [N]} = \left\lceil \frac{\text{Old Payload Size}}{\text{New Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{4384 \text{ bytes}}{1480 \text{ bytes}} \right\rceil$$

$$N = \lceil 2.96 \rceil$$

$$= 3$$

Total length of the last IP fragment =

$$[\text{HLEN} * 4] + [\text{Old Payload Size} - (N-1) * \text{New Payload Size}] \text{ bytes}$$

$$= 20 \text{ bytes} + [4384 - (3-1) * 1480] \text{ bytes}$$

$$= [20 + 1424] \text{ bytes}$$

$$= 1444 \text{ bytes}$$

[MCQ]

[GATE-CS-2004] [2 Mark]



#Q. Consider three IP networks A, B and C. Host H_A in network A sends messages each containing 180 bytes of application data to a host H_C in network C. The TCP layer prefixes a 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network is : [MTU]

A : 1000 bytes

B : 100 bytes

C : 1000 bytes

Assuming that the packets are correctly delivered, how many bytes, including headers, are delivered to the IP layer at the destination for one application message, in the best case ? [Consider only data packets.]

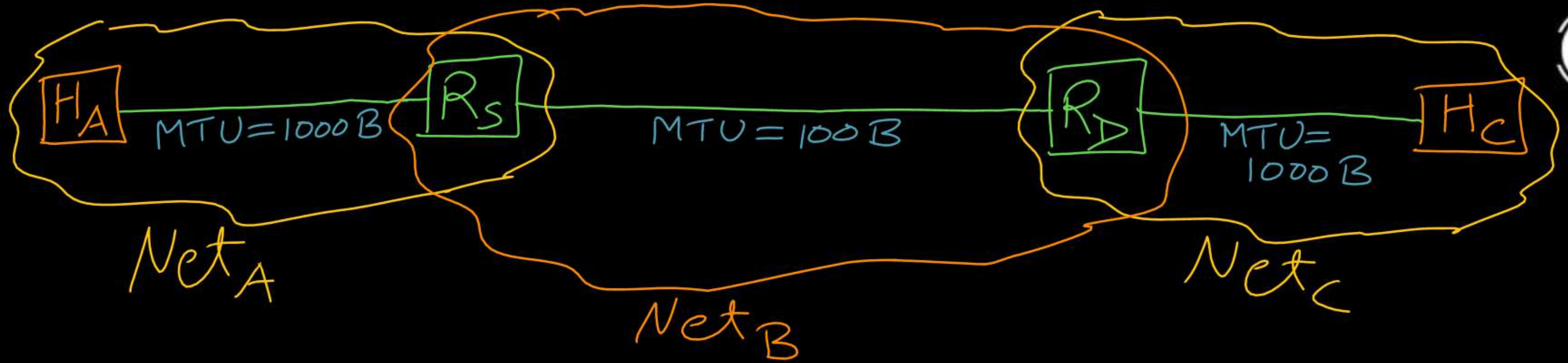
A 200

B 220

C 240

✓ D 260

Ans: D



TCP Header Size = [20 bytes]

TCP Payload Size = [180 bytes]

TCP Segment Size = 200 bytes

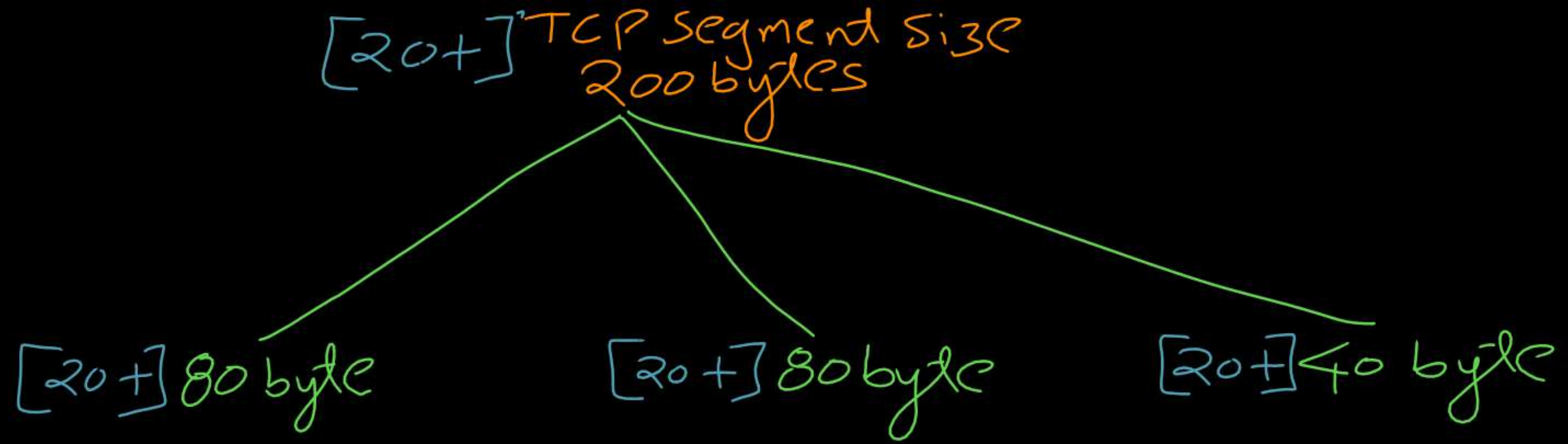
MTU for network B = [100 bytes]

IPv4 Header Size = [20 bytes]

Maximum Payload Size = [MTU - Header Size] bytes

= [100 - 20] bytes

= 80 bytes



$$\begin{aligned} \text{Ans} &= (100 + 100 + 60) \text{ bytes} \\ &= 260 \text{ bytes} \end{aligned}$$

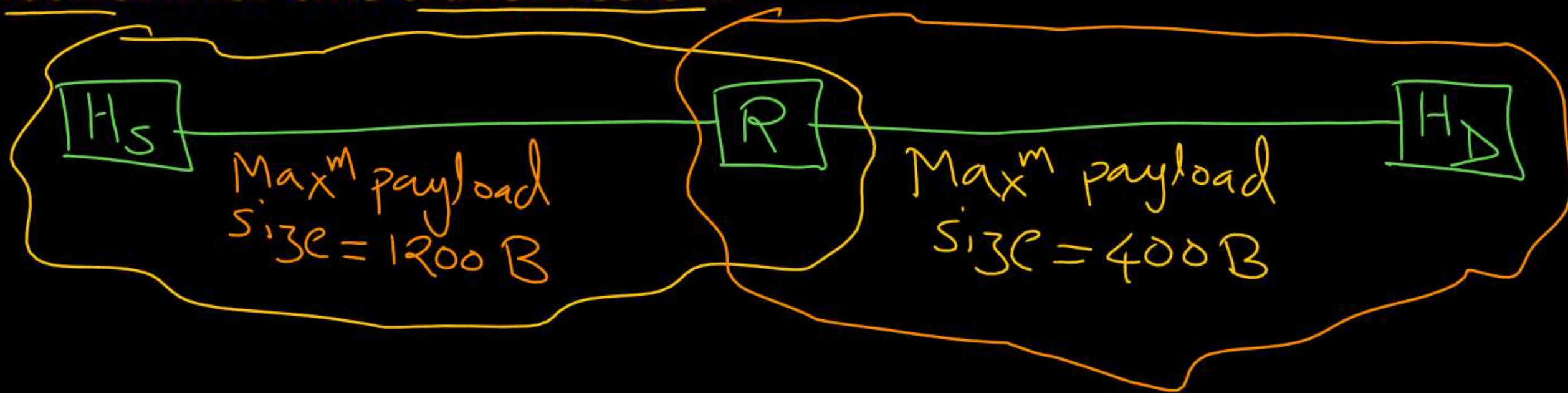
[MCQ]

[GATE-IT-2004] [2 Mark]



#Q. A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per datagram and the second network can carry a maximum payload of 400 bytes per datagram, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

- A 40 bytes
- B 80 bytes
- ☒ C 120 bytes
- D 160 bytes



Ans: C

TCP Segment size
2100 bytes



$$\boxed{\text{Ans} = 6 \times 20 \text{ byte} = 120 \text{ byte}}$$

Example 9 :-

[NAT]



#Q. Consider UDP segment of size 1000 bytes is passed to IPv4 protocol for delivery. MTU for source network is 400 bytes and IPv4 header size is 20 bytes. Calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation?

Ans = 3



MTU = [400 bytes]

IPv4 Header Size = [20 bytes]

$$\begin{aligned}\text{Maximum Payload Size} &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (400 - 20) \text{ bytes} \\ &= 380 \text{ bytes}\end{aligned}$$

* Payload size for the fragments where M bit is one, should be in words of 8 bytes.

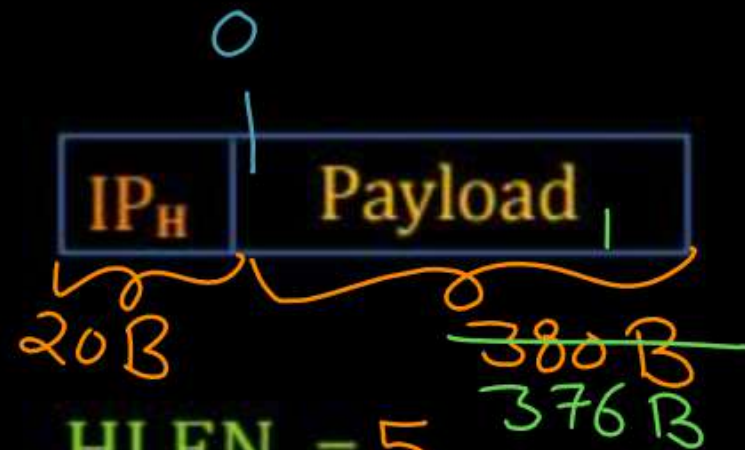
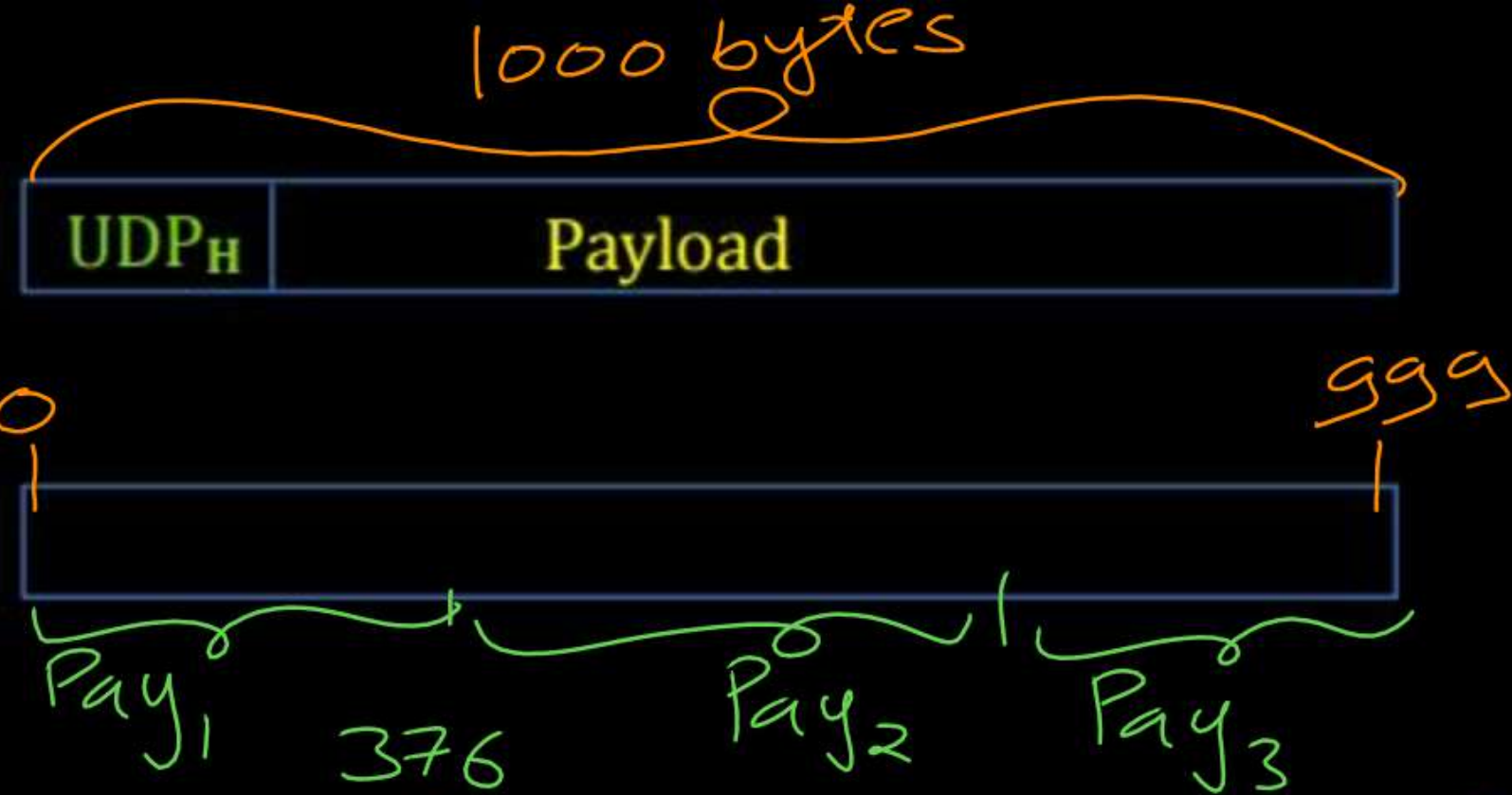
$$\begin{aligned}&[\text{Payload size should be multiple of 8 bytes}] \\ &= 376 \text{ bytes}\end{aligned}$$

$$\text{UDP Segment Size} = 1000 \text{ bytes}$$

$$\text{Total Number of IP fragments [N]} = \left\lceil \frac{\text{UDP Segment Size}}{\text{Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{1000 \text{ bytes}}{380 \text{ bytes}} \right\rceil = \left\lceil \frac{1000 \text{ bytes}}{376 \text{ bytes}} \right\rceil$$

$$N = 3$$



$$\text{HLEN} = 5$$

$$\text{TL} = 396$$

$$\text{Offset} = \frac{0}{8} = 0$$

$$\text{MF bit} = 1$$

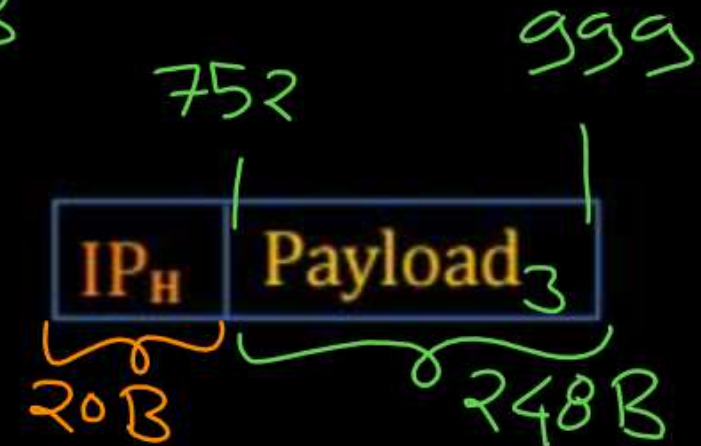


$$\text{HLEN} = 5$$

$$\text{TL} = 396$$

$$\text{Offset} = \frac{380}{8} = 47$$

$$\text{MF bit} = 1$$



$$\text{HLEN} = 5$$

$$\text{TL} = 268$$

$$\text{Offset} = \frac{752}{8} = 94$$

$$\text{MF bit} = 0$$

[NAT] (H.W.)

11T-G

[GATE-2018][2 Mark]



#Q. Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

[NAT]

H.W.

11SC

[GATE-2024][2 Mark]



#Q. Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____.

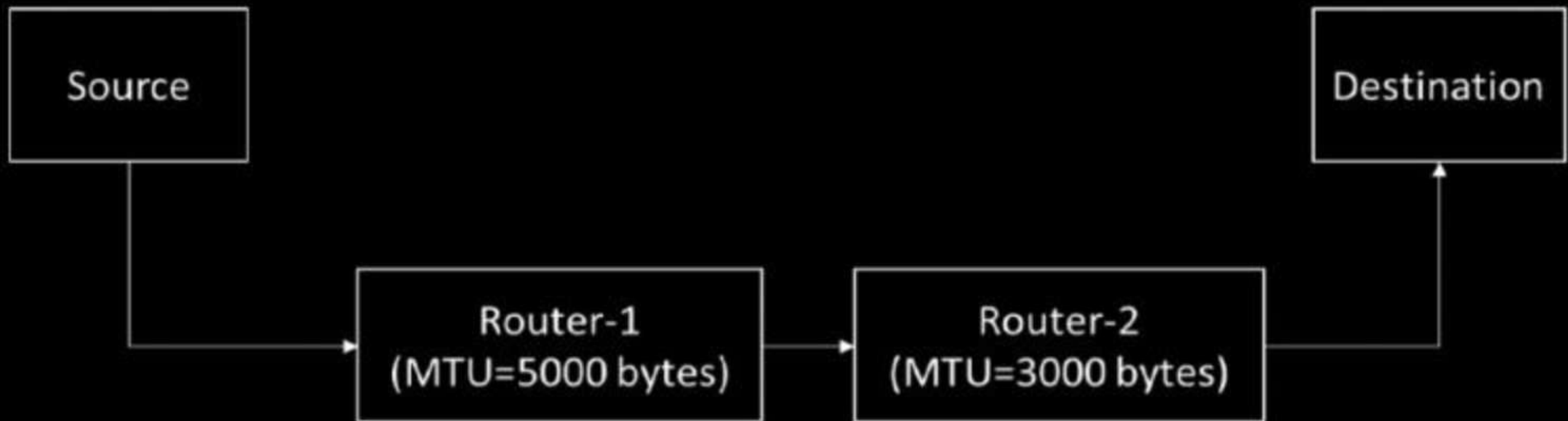
[NAT] / H.W.

IIT-R

[GATE-2025][2 Mark]



#Q. Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____. (Answer in integer)





2 mins Summary



Topic

Fragmentation & Reassembly



THANK - YOU

